

OOCAA Document Management System (DMS)

Full Technical & Financial Proposal



Prepared for:
Oromia Occupational Competency Assurance
Agency (OOCAA)



Prepared by:
Future Software Solutions(FSS)

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1. Executive Summary

OOCAA seeks to modernize and digitize its document handling processes across headquarters and branch offices. The proposed Document Management System (DMS) will provide a centralized, secure, scalable, and role-based digital repository for all institutional documents.

The solution will streamline document registration, classification, tracking, approval workflows, metadata management, audit trails, and automated reporting. The system will be hosted on a secure cloud or the National Data Center and fully integrated with OOCAA's current network infrastructure.

This proposal outlines the technical requirements, project methodology, modules, timelines, and cost breakdown budget.

2. Introduction

The OOCAA Document Management System (DMS) is designed to eliminate manual document handling processes and replace them with a fully automated, searchable, trackable, and secure digital archive system.

It improves:

- Access to documents
- Document lifecycle management
- Transparency
- Institutional efficiency
- Data accuracy and security

3. Project Objectives

- Develop a secure, scalable web-based DMS accessible across OOCAA and all branch offices.
- Digitize and centralize document storage and metadata management.
- Automate document workflows, tracking, and approvals.

- Provide role-based access control with full audit logs.
- Support multilingual interface: Afaan Oromo, Amharic, and English.
- Provide real-time dashboards, performance metrics, and reporting tools.
- Deploy the system to cloud/server infrastructure with user training and support.

4. Scope of Work

The project scope includes:

- Requirement gathering & analysis
- System design (UI/UX, architecture, data model)
- Development of full DMS modules
- Integration with OOCAA infrastructure
- API development for future third-party integrations
- Testing (unit, integration, UAT, performance, security)
- Deployment & configuration
- Documentation (technical, user, admin manuals)
- Training (technical & end-user)
- 1-year maintenance free support

5. System Overview

The DMS will allow OOCAA to:

- Register, store, and categorize documents
- Perform full-text and metadata-based searches
- Track document access, edits, sharing, and ownership
- Manage workflows and approvals
- Generate reports on document usage, performance, and operations

6. Modules of the Document Management System

6.1 Core Modules

- User & Role Management Module
- Document Registration & Metadata Management

- Document Storage & Archiving
- Document Tracking Module
- Advanced Search & Filtering
- Audit Trails & Activity Logs
- Version Control Module

6.2 Operational & Support Modules

- Dashboard & Analytics (Graph + Tabular Reports)
- Notification & Alert System (Email / In-app)
- Role-Based Access Control (RBAC)
- Branch Management Module
- Backup & Disaster Recovery Module
- API Integration Module
- Reporting Module (User reports, Access reports, Audit reports)

7. Functional Requirements

7.1 Document Metadata Management

- Register new documents with full metadata fields
- Edit/update metadata
- Assign unique document IDs
- Categorize documents by type, office, classification level

7.2 Document Tracking

- Track document movement digitally
- Request access from owning office
- Approve/reject document access
- Track each user's interaction history

7.3 Search

- Search by metadata fields
- Full-text search
- Filter by branch, office, department, date, classification

7.4 Reporting

- Reports by user, document type, branch
- Usage trend reports
- Monthly/quarterly/yearly document logs

8. Non-Functional Requirements (Quality Attributes)

8.1 Usability

- Simple, multilingual, responsive UI
- Accessible on desktop, tablet, mobile

8.2 Performance

- Response time < 5 seconds for major functions
- Optimized database queries

8.3 Security

- Robust authentication & authorization
- Encrypted data storage
- 2FA (optional)
- Full activity logging

8.4 Availability

- 99.99% uptime
- Deployed on OOCAA Server or cloud or national data center

8.5 Scalability

- Vertical and horizontal scaling supported

9. System Architecture

9.1 Layers

- Presentation Layer (Web/Mobile Responsive)
- Application Layer (API-based services)
- Business Logic Layer
- Data Access Layer
- Database Layer (PostgreSQL)

9.2 Technology Stack

- Frontend: React
- Backend: Laravel
- Database: PostgreSQL
- Server: Linux-based cloud server
- Security: SSL, TLS, JWT

10. External Interface Requirements

10.1 User Interfaces

- Browser-based
- Accessible on LAN or Woredanet VPN and cloud

10.2 Communications Protocols

- HTTP/HTTPS
- TCP/IP

11. Methodology – Agile

Development, deployment, training will follow **4 Sprints (1 weeks each)**

Includes: requirement validation → design → development → testing → deployment.

12. Organization & Team Composition

12.1 Project Governance Structure

- Project Steering Committee (OOCAA + Vendor Senior Management)
- Project Manager (Vendor)
- Technical Lead

- Business Analyst / System Analyst
- Backend Developers
- Frontend Developers
- QA & Testing Team
- UI/UX Designer
- Deployment & DevOps Engineer
- Documentation & Training Team

12.2 Team Composition Table

Role	Quantity	Responsibility
Project Manager	1	Overall control & coordination
Technical Lead	1	Oversees system architecture & code quality
Backend Developers	1	API, logic, database
Frontend Developer	1	UI/UX, web interface
DevOps Engineer	1	Deployment, cloud, security
Business/Systems Analyst ,Trainer & Documentation Officer	1	User manuals, training
Total Staff	6	

13. Resource Requirements

13.1 Hardware

- OOCOA Server /Cloud server / National Data Center VM
- Backup storage
- Network infrastructure

13.2 Software

- Database server
- Application server
- Backup system

14. Risk Assessment & Mitigation

Risk	Impact	Mitigation
Requirement changes	Medium	Agile methodology & sprint review
Server downtime	High	Cloud redundancy & backup
User adoption resistance	Medium	Training & onboarding
Security threats	High	SSL, encryption, penetration testing

15. Deliverables

1. Fully functional OOCAA DMS
2. SRS, architecture, and design documents
3. User manuals & admin manuals
4. Deployment & configuration files
5. Source code
6. Training completion report
7. Maintenance plan

16. Training Plan

Training Type	Participants	Duration	Description
Technical Training	7 IT Staff	6 Days	Backend admin, server, troubleshooting
User Training	60 Users	4 Days	System usage, workflows, reports

17. Maintenance & Support (1 Year)

- Bug fixing
- Security updates
- Server monitoring
- Backup and restore support
- Minor feature updates

18. Project Timeline (1 Months)

Task	Week 1	Week 2	Week 3	Week 4
Requirement Gathering	✓			
Analysis	✓			
System Design	✓	✓		
Development		✓	✓	
Testing		✓	✓	
Deployment			✓	✓
Training and hand over		✓	✓	✓

19. Financial Proposal

Cost Breakdown

Category	Amount (ETB)
System Development (Backend + Frontend + Architecture)	2,000,000
System Analysis, Design & Documentation	450,000
Project Management	300,000
UI/UX Design	150,000
Testing & QA	300,000
Deployment & Cloud Configuration	250,000
Training (Users + Technical)	250,000
Maintenance & Support (1 Year)	700,000
Infrastructure (Server, Backup, Security Configurations)	350,000
Contingency (10%)	250,000
Total Estimated Cost	5,000,000 ETB

1. Company Profiles

1.1 CV Project Manager and Full Stack Developer

Full Name: Dejene Kasa Alemu

Phone: +251-927-25-67-24,

Email: dejenekasa1@gmail.com

Professional Summary

Results-driven Project Manager with experience overseeing full-cycle software development projects. Skilled in planning, resource allocation, Agile/Scrum, client communication, and ensuring timely, high-quality project delivery.

Core Competencies

- Project Planning & Scheduling
- Agile/Scrum Methodologies
- Risk Management
- Budget Management
- Team Coordination
- Client & Stakeholder Communication
- Quality Assurance
- Documentation & Reporting

Professional Experience

Project Manager:

Adama City Administration Science and Technology Office *08/2024 - 08/2025*

Full Stack Software Developer:

Oromia Science and Technology Authority *07/20219 – 08/2024*

Project Manager – Future Software Solutions *04/09/2025 – Present*

- Lead and manage development teams for web and software projects.
- Prepare project schedules, assign tasks, and monitor progress.
- Ensure project timelines, scope, and budget are met.
- Communicate project updates to clients and senior management.
- Manage risks and implement mitigation strategies.

- Prepare technical and progress reports.

Education

Bachelor's Degree in Software Engineering

Adama Science and Technology University, 2017

Certifications

- Agile Project Management
- Acknowledgement of Different project completion and company recognitions

Technical Skills

- Jira, Trello
- MS Project
- Git / GitHub
- Figma

Languages

- English – Excellent
- Afan Oromo – Excellent
- Amharic – Excellent

Certificates from different Organizations

1. Adama City Integrated Document Management System (IDMS)
from LT ICT Solution



2. Oromia State University Property Management System



3. Oromia Science and Technology Authority (ASTO DMS)



4. Oromia Green Legacy Reporting System



1.2 Training completion Certificates

1. Virtualizations



2. SMS Center Configurations



1.2. CV – Full Stack Developer

Full Name: Temam Aman

Address: Adama City

Phone: +251 97 387 0819

Email: temamaman156@gmail.com

Professional Summary

Full Stack Developer with strong expertise in both front-end and back-end development. Experienced in building scalable, secure, and user-friendly applications using modern frameworks.

Core Competencies

- Front-End: React, Next.js, HTML, CSS, JavaScript
- Back-End: Node.js, Express, PHP, Laravel
- Database: PostgreSQL, MySQL, MongoDB
- REST API Development
- Authentication (JWT, OAuth)
- Git, CI/CD
- System Design

Professional Experience

Full Stack Developer – **Guba Net ICT Solution**

Oct/2022 – Present

Full Stack Developer – **Future Software Solutions**

Oct/2025 – Present

- Develop full-stack applications with React, PostgreSQL, and Node.js.
- Build responsive UI components and reusable code modules.
- Integrate APIs, authentication, and server-side business logic.
- Optimize system performance and security.
- Collaborate with project managers and other developers.

Education

Bachelor's Degree in Electrical and Computer Science Engineering
Samara University, 2021

Projects

- Customer Feedback System (React + Express + PostgreSQL)
- Admin Dashboard with Authentication
- E-commerce Back-End API
- Mobile-responsive Frontend Systems

Languages

- English – Excellent
- Afan Oromo – Excellent
- Amharic – Excellent



TEMAM AMAN KABETO

■ sisinineaman.com@gmail.
com
■ 0986695595
■ Addis Ababa

Objective

As a recent graduate, I am seeking a role which allows me to continue learning and perfecting my skills as I provide high-quality work, and encourages me to flourish as a Computer Engineer.

Skills

Soft Skill

- ✓ Communication Skill
- ✓ Problem Solving Skill
- ✓ Organizational Skill

Hard Skill

- ✓ Computer Skill
- ✓ leadership Skill
- ✓ software development skill

Reference

JEMAL Mohammed

Department head and lecture

At Samara University

jemalmoha35@gmail.co

+251 9143 52253

Rajat kapila - "Samara

Assistant professor

rajatkapila72@gmail.com

+251 939551214

Ankamma Rao J - "Samara
university"

Assistant professor (IEEE

Member) and Lecture at

samara university

+251 93 7602833

Education

Samara University

CGPA 3.78

*Graduate
On year
2021*

Projects

Land Information Management System

I fully participate when Adama City LIMS is developed by GubaNet ICT Solution Plc.

Experience

- Over 3 years of extensive experience in designing and developing different types of application ranging from web application to specialized business applications and ERP systems.
- Experience in complete SDLC phases of project involving analysis, architecture, design, development, testing,
- proposal or term of reference documentation, business requirement specification, system requirementspecification and high-level object design documentation,
- Deploy various Client-Server, Web and Windows Applications.
- Experience in analysis and creation of Web applications using Web form, MVC framework, Java script, Ajax,jQuery.
- Experience in analysis and creation of Single Page applications with dot net Core and Angular framework.
- Experience in Microsoft SQL Server (2008, 2012, 2014, 2017, 2019) database
- Responsible in designing and developing data migration tools depending on existing client data.

I have fully participated when the following systems are developed by Guba Net ICT Solution as a software developer.

Client..... RADA Collage

Project..... **E-learning system and Student Information Management System (SIMS)**

Role..... Application Developer

Description..... Working as a par time full stack developer, for Online Learning system

Date..... Oct-2021 Mar-2022

Responsibilities:

- ✓ Responsible for to formulate Proposal, requirement analysis document (RAD), System specification documents (SRS), Design and build a particular model of proposed system.
 - ✓ Responsible for deploy on remote IIS server
 - ✓ Responsible for the front end part of the project
-

Client..... Super Market

Project.....**Bussiness ERP including inventory and Customer Management System**

Role..... **Application Developer**

Description... Working as a Full time full stack developer, for **Bussiness ERP including inventory** system

Date..... 02/10/ 2021 to 29/12//2021

Responsibilities:

- ✓ Responsible for designing the Back end of application.
- ✓ Responsible for to formulate Proposal, requirement analysis document (RAD), System specification documents (SRS), Design and build a particular model of proposed system.
- ✓ Responsible for deploy on remote IIS server
- ✓ Responsible for the front end part of the project

Client:.....Select collage

Project:.....**Student registration System**

Description:.....Responsible for Designing, developing and organizing implementation
Of the HR Module of the system.

Role:.....**Software Developer**

Responsibilities:

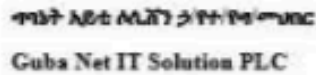
I have participated in the end-to-end development process of the company's Student registration System
Responsible for the designing user interface of the system
Responsible for the deployment of the system on their
server.

Client:.....**Adama Land Adminstration**

Project:..... Land Management system

Description Responsible for Designing, developing and organizing implementation
Of the LIMS Module at Guba Net.

Role:**Software Developer**



30/02/2017

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1.3. CV – Document Analyst

Full Name: **Tesfaye Mulisa**

Phone: **251- 912-900-461**

Email: **mulisabekan@gmail.com**

Professional Summary

Dynamic and detail-oriented Full Stack Developer with hands-on experience designing, developing, and deploying high-quality web applications. Strong problem-solving ability and teamwork skills.

Core Skills

- React.js, Vue.js (optional), JavaScript ES6
- Laravel, Node.js, Express
- RESTful APIs
- Database Design (PostgreSQL, MySQL)
- Version Control (Git/GitHub)
- Deployment (Docker, VPS, cPanel)
- Unit Testing

Professional Experience

Full Stack Developer – Oromia Finance Bureau

Sep-2022– *Present*

- Create modular and scalable web applications.
- Write clean, maintainable, and testable code.
- Implement secure authentication workflows.
- Maintain documentation and provide technical support.
- Participate in sprint planning and code reviews.

Education

Msc in Computer Science

Certifications

- Full Stack Development Certification
- JavaScript / React Certification

4. CV – Document Analyst

Full Name: Musa Ahmed

Phone: +251 91 921 5261

Email: musaahmed@gmail.com

Professional Summary

Document Analyst with strong experience in organizing, reviewing, validating, and processing technical and administrative documentation. Skilled in ensuring accuracy, compliance, and proper version control.

Key Competencies

- Document Review & Validation
- Data Entry & Accuracy Checking
- Quality Assurance
- Report Preparation
- Version Control & Archiving
- MS Word, Excel, PDF Tools
- Attention to Detail
- Communication & Coordination

Professional Experience

Document Analyst – Adama City Administration

06/2024 – Present

Document Analyst – Future Software Solution (FSS)

06/2025 – Present

- Review and organize project and administrative documents.
- Ensure all documents meet compliance and format standards.
- Extract key data and prepare summary reports.
- Coordinate with developers, HR, and project managers.
- Manage digital and physical document archives.

Education

BA in Management / Information Systems / Related Field
Wolkite University, 2017

Technical Skills

- Microsoft Office Suite
- Google Workspace
- Document Collaboration Tools
- PDF Editing Tools

5. CV – Human Resource Officer (HR)

Full Name: Madanit Teshome

Phone: 0949916686

Email: madanitt@gmail.com

Professional Summary

Human Resource Officer with experience in recruitment, employee relations, onboarding, performance management, and HR documentation. Strong communication, ethical conduct, and organizational skills.

Core Competencies

- Recruitment & Selection
- Employee Onboarding
- HR Policies & Procedures
- Payroll Coordination
- Leave & Attendance Management
- Conflict Resolution
- Interview Scheduling
- HR Reporting
- Training Coordination

Professional Experience

HR Officer – Future Software Solutions (FSS)

07/2025 – Present

- Manage recruitment and selection processes.
- Maintain HR documents, contracts, and employee records.
- Support employee relations and handle grievances.

- Prepare HR reports for management.
- Coordinate performance evaluations and training.

Education

BA in Human Resource Management / Business Administration
Harambe Univeristy Collegue, 2022

Technical Skills

- HR Software
- Microsoft Office
- Email & Communication Tools

21. Conclusion

The OOCOA Document Management System will fundamentally transform document handling across all branches by providing a centralized, secure, transparent, and efficient digital environment. The proposed system aligns with OOCOA's operational goals and government digital transformation initiatives.

This proposal demonstrates the technical, operational, and financial feasibility of the project, ensuring an optimized system delivered within the allocated **5 million ETB budget** and 1-month timeframe.